Week 2: Structured vs. Unstructured Data

- Classification of data based on it format and schema rules.
- Structured Data:
    - Has fixed schema and fits into rows and columns
- Unstructured Data:
    - Has no fixed schema and can have more complex format

Key Areas	Structured Data	Unstructured Data
Format	Strict, predefined data model.	No predefined format.
Storage	Have rigid schemas, like in relational databases.	Stores in its naïve format in nonrelational databases.
Use Cases	Machine Learning and Machine Learning Algorithms	Natural Language Processing and a rich and diverse data source for Gen AI models.
Complexity	Easier to manipulate and analyze for general business users with traditional tools.	More complex and requires specialized skills and tools to parse and analyze.

Structured Data

- This can include both quantitative data and qualitative data.
- For example, a financial report with company names, expense values and reporting periods organized into rows and columns is considered structured data.

How is Structured Data used?

- Stored in tabular formats, like Excel, spreadsheets and relational databases.
- By using Structured Query Language (SQL) users can efficiently niput, search and manipulate structured data within RDBMS.
- Use cases :
    - AI model training
    - Customer Relationship Management (CRM)
    - Business Intelligence (BI)
    - Inventory Management
    - Search Engine Optimization (SEO)
- ***Pros & Cons***
    - ***Pros:***
    - **Works well with machine learning**
    - **Accessible and easy to use**
    - **Abundance of tools**
    - ***Cons:***
    - **Limited usage**
    - **Limited storage options**

Unstructured Data

- Large datasets and comprise 90% of all enterprise-generated data.
- The high volume is due to the emergence of big data-the massive, complex datasets from the internet and other connected technologies.
- This data can be both textual and non-textual and both qualitative and quantitative.

How is Structured Data used?

- It is best managed in nonrelational or NoSQL databases or in data lakes, which are designed to handle massive amounts of raw data in any format.
- Often, machine learning, advanced analytics, and natural language processing (NLP) are used to extract valuable insights from unstructured data.
- Use cases :
    - RAG
    - GenAI
    - Customer Behavior and Sentiment Analysis
    - Predictive Data Analytics
    - Chatbot text analysis
- ***Pros & Cons***
    - ***Pros:***
    - **Flexibility**
    - **Fast accumulation rates**
    - **Easy and cheap to store**
    - ***Cons:***
    - **Requires expertise**
    - **Specialized tools**
    - **Data cleanliness**

AI and Unstructured Data Analytics

- AI can quickly process large volumes of data.
- With machine learning and natural language processing (NLP), AI algorithms can sift through unstructured data to find patterns and make real-time predictions or recommendations.

Semi-Structured Data

- Semi-structured data is the “bridge” between structured and unstructured data. It is useful for web scraping and data integration**.**
- They do not have a predefined data model.
- But uses metadata to identify specific data characteristics and scale data into recorded and preset fields.
- Metadata enables semi-structured data to be better catalogued, searched, and analyzed.
- Examples of semi-structured data include JavaScript Object Notation (JSON), comma-separated values (CSV), and eXtensible Markup Language (XML) files.